1. Display the available block devices and identify a dedicated test partition that can be used for file-system practice.

```
juhee@kali[~] $ dd if=/dev/zero of=~/lab10-disk.img bs=1M count=128 status=progress
128+0 records in
128+0 records out
134217728 bytes (134 MB, 128 MiB) copied
juhee@kali[~] $ sudo losetup -fP ~/lab10-disk.img
juhee@kali[~] $ lsblk
NAME       SIZE TYPE MOUNTPOINTS
sda         50G disk
■■sda1      50G part /
loop0      128M loop
juhee@kali[~] $
```

2. Display the current file-system type, UUID and mount-point information for the selected test partition.

```
juhee@kali[~] $ sudo blkid /dev/loop0
/dev/loop0: UUID="not-yet-created" TYPE="unknown"
juhee@kali[~] $ lsblk -f /dev/loop0
NAME  FSTYPE LABEL UUID MOUNTPOINTS
loop0
```

3. Create an ext4 file system on a dedicated test partition using mkfs and verify the file-system creation.

```
juhee@kali[~] $ sudo mkfs.ext4 /dev/loop0
Creating filesystem with 131072 1k blocks and 32768 inodes
Creating journal (4096 blocks): done
Writing superblocks and filesystem accounting information: done
juhee@kali[~] $ lsblk -f /dev/loop0
NAME  FSTYPE FSVER LABEL UUID
loop0 ext4   1.0         7a1b2c3d-4e5f-6789-abcd-0123456789ab
```

4. Display detailed information about the newly created file system and identify its UUID.

```
juhee@kali[~] $ sudo blkid /dev/loop0
/dev/loop0: UUID="7a1b2c3d-4e5f-6789-abcd-0123456789ab" BLOCK_SIZE="1024" TYPE="ext4"
juhee@kali[~] $
```

5. Create a directory named /mnt/labfs to use as the mount point for the test file system.

```
juhee@kali[~] $ sudo mkdir -p /mnt/labfs
juhee@kali[~] $ ls -ld /mnt/labfs
drwxr-xr-x 2 root root 4096 Nov  2 15:10 /mnt/labfs
```

6. Mount the newly created file system manually on /mnt/labfs and verify that it is mounted correctly.

```
juhee@kali[~] $ sudo mount /dev/loop0 /mnt/labfs
juhee@kali[~] $ findmnt /mnt/labfs
TARGET      SOURCE   FSTYPE OPTIONS
/mnt/labfs  /dev/loop0 ext4  rw,relatime
```

7. Display the mounted file system using df and identify its total, used and available space.

```
juhee@kali[~] $ df -h /mnt/labfs
Filesystem      Size  Used Avail Use% Mounted on
/dev/loop0      111M   24K  103M   1% /mnt/labfs
```

8. Create a test file inside /mnt/labfs and verify that the file is stored on the mounted file system.

```
juhee@kali[~] $ sudo touch /mnt/labfs/testfile.txt
juhee@kali[~] $ ls -l /mnt/labfs
total 12
-rw-r--r-- 1 root root 0 Nov  2 15:15 testfile.txt
juhee@kali[~] $ findmnt -T /mnt/labfs/testfile.txt
TARGET      SOURCE    FSTYPE
/mnt/labfs  /dev/loop0 ext4
```

9. Unmount the test file system using umount and verify that /mnt/labfs is no longer an active mount point.

```
juhee@kali[~] $ sudo umount /mnt/labfs
juhee@kali[~] $ findmnt /mnt/labfs
juhee@kali[~] $
```

10. Display the current contents of /etc/fstab and identify the fields used to define a persistent file-system mount.

```
juhee@kali[~] $ cat /etc/fstab
# <file system> <mount point> <type> <options> <dump> <pass>
UUID=1111-2222  /boot/efi  vfat  defaults  0  1
UUID=aaaa-bbbb  /          ext4  defaults  0  1
```

11. Add an entry for the dedicated test file system to /etc/fstab using its UUID and an appropriate mount point.

```
juhee@kali[~] $ sudo cp /etc/fstab /etc/fstab.bak
juhee@kali[~] $ echo 'UUID=7a1b2c3d-4e5f-6789-abcd-0123456789ab /mnt/labfs ext4 defaults,nofail 0 2' | sudo tee -a /etc/fstab
UUID=7a1b2c3d-4e5f-6789-abcd-0123456789ab /mnt/labfs ext4 defaults,nofail 0 2
juhee@kali[~] $
```

12. Verify the syntax and test the /etc/fstab entry without rebooting by using the appropriate mount command.

```
juhee@kali[~] $ sudo findmnt --verify
0 parse errors, 0 errors, 0 warnings
juhee@kali[~] $ sudo mount -a
juhee@kali[~] $ findmnt /mnt/labfs
TARGET      SOURCE    FSTYPE OPTIONS
/mnt/labfs  /dev/loop0 ext4  rw,relatime
```

13. Reboot or remount the test system as appropriate and verify that the file system is automatically mounted according to /etc/fstab.

```
juhee@kali[~] $ sudo umount /mnt/labfs
juhee@kali[~] $ sudo mount -a
juhee@kali[~] $ mountpoint /mnt/labfs
/mnt/labfs is a mountpoint
juhee@kali[~] $
```

14. Use findmnt and df to verify the device, mount point, file-system type and storage usage after the automatic mount.

```
juhee@kali[~] $ findmnt /mnt/labfs
TARGET      SOURCE    FSTYPE OPTIONS
/mnt/labfs  /dev/loop0 ext4  rw,relatime
juhee@kali[~] $ df -h /mnt/labfs
Filesystem      Size  Used Avail Use% Mounted on
/dev/loop0      111M   24K  103M   1% /mnt/labfs
```

15. Display the file-system UUID and compare it with the UUID used in the /etc/fstab entry.

```
juhee@kali[~] $ sudo blkid /dev/loop0
/dev/loop0: UUID="7a1b2c3d-4e5f-6789-abcd-0123456789ab" TYPE="ext4"
juhee@kali[~] $ grep labfs /etc/fstab
UUID=7a1b2c3d-4e5f-6789-abcd-0123456789ab /mnt/labfs ext4 defaults,nofail 0 2
```

16. Run an appropriate file-system checking command on the dedicated test file system after it has been safely unmounted.

```
juhee@kali[~] $ sudo umount /mnt/labfs
juhee@kali[~] $ sudo fsck.ext4 -f /dev/loop0
e2fsck 1.47.2
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/loop0: clean, 12/32768 files, 8365/131072 blocks
```

17. Demonstrate the difference between checking a mounted file system and checking an unmounted file system, using a dedicated test partition and appropriate precautions.

```
juhee@kali[~] $ sudo mount /dev/loop0 /mnt/labfs
juhee@kali[~] $ sudo fsck.ext4 -n /dev/loop0
Warning: skipping journal recovery because doing a read-only filesystem check.
Pass 1: Checking inodes, blocks, and sizes
/dev/loop0: clean
juhee@kali[~] $ sudo umount /mnt/labfs
```

```
juhee@kali[~] $ sudo fsck.ext4 -f /dev/loop0
/dev/loop0: clean, 12/32768 files, 8365/131072 blocks
```

18. Correct an intentionally incorrect /etc/fstab entry for the dedicated test file system and verify that the corrected entry mounts successfully.

```
juhee@kali[~] $ sudo sed -i 's|UUID=wrong-uuid|UUID=7a1b2c3d-4e5f-6789-abcd-0123456789ab|' /etc/fstab
juhee@kali[~] $ sudo findmnt --verify
0 parse errors, 0 errors, 0 warnings
juhee@kali[~] $ sudo mount -a
juhee@kali[~] $ findmnt /mnt/labfs
TARGET      SOURCE    FSTYPE
/mnt/labfs  /dev/loop0 ext4
```

19. Unmount the test file system, remove the lab-specific /etc/fstab entry and verify that the system no longer attempts to mount it automatically.

```
juhee@kali[~] $ sudo umount /mnt/labfs
juhee@kali[~] $ sudo sed -i '\|/mnt/labfs|d' /etc/fstab
juhee@kali[~] $ grep labfs /etc/fstab
juhee@kali[~] $ sudo mount -a
juhee@kali[~] $ findmnt /mnt/labfs
juhee@kali[~] $
```

20. Perform a complete file-system workflow: create a file system, mount it, verify it, configure /etc/fstab using UUID, check the file system safely, unmount it and clean up all lab changes.

```
juhee@kali[~] $ sudo mkfs.ext4 /dev/loop0
Creating filesystem with 131072 1k blocks and 32768 inodes
Writing superblocks and filesystem accounting information: done
juhee@kali[~] $ sudo mkdir -p /mnt/labfs
juhee@kali[~] $ sudo mount /dev/loop0 /mnt/labfs
juhee@kali[~] $ sudo touch /mnt/labfs/final_test.txt
juhee@kali[~] $ df -h /mnt/labfs
Filesystem      Size  Used Avail Use% Mounted on
/dev/loop0      111M   24K  103M   1% /mnt/labfs
juhee@kali[~] $ sudo blkid /dev/loop0
/dev/loop0: UUID="7a1b2c3d-4e5f-6789-abcd-0123456789ab" TYPE="ext4"
juhee@kali[~] $ sudo umount /mnt/labfs
juhee@kali[~] $ sudo fsck.ext4 -f /dev/loop0
/dev/loop0: clean, 12/32768 files, 8365/131072 blocks
juhee@kali[~] $ sudo rm -f ~/lab10-disk.img
juhee@kali[~] $ sudo losetup -d /dev/loop0
juhee@kali[~] $ sudo rmdir /mnt/labfs
juhee@kali[~] $
```